

ESC101

Minimizing the Residue Remaining in Moisturizer Pump Dispensers

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1. About the opportunity

This design report recommends the design of a pump dispenser with a concave bottom that can be transformed into a jar to address the issue of inaccessible residue at the bottom of moisturizer bottles [Figure-1], [Figure-2]. Though other moisturizer dispensers exist, designs such as tubes often suffer from volume limitations and squeeze bottles require additional effort to operate in comparison to conventional pump bottles [1], [2]. Therefore, this report recommends a design to minimize wastage in moisturizer pump bottles. The document will describe how this design addresses the opportunity of moisturizer bottle residue, the requirements it fulfills, as well as comparisons made between the recommended design, a reference design, and the four alternatives explored: a crater, an upside-down bottle holder, a cartridge system, and a one-way valve.



Figure-1: A cut-open moisturizer pump bottle. A lot of moisturizer can be seen in locations inaccessible by the pump.



Figure-2: Moisturizer extracted from a cut-open bottle.

2. Stakeholders

The primary stakeholders of this design report are cost-conscious first-year students who live in residence or rent. Approximately half of first-year engineering students live in residence [3], therefore, their housing situations are non-permanent. Accordingly, solutions designed should aim to avoid causing permanent changes to their surroundings while maximizing the amount of product extracted

3. NGO, Requirements, and Evaluation Criteria

The requirements found in 3.2 were adapted from the NGOs found in Appendix-B. Justification for each requirement can be found in Appendix-C.

The Evaluation criteria were developed in accordance with measurable requirements that allow for comparison between designs.

3.1. Need Statement

We need a mechanism that will allow users to access most of the container's contents without requiring any permanent installation.

3.2. Requirements and Evaluation Criteria

Table-1: Goals, Requirements, and Evaluation Criteria

Goals	Requirements	Evaluation Criteria
1: Functionality The design should dispense a majority of the moisturizer in a timely manner, while requiring minimal effort to use.	1: The dispensing mechanism should be able to extract $\geq 88.31\%$ of the product by volume.	1: The greater the percentage volume of product the design can extract, the better
	2: No more than 30N of force should be required to operate the design per dispensation cycle.	2: The smaller the force in newtons required to operate the design, the better.
	3: The design should take no longer than 5 seconds to operate	3: The shorter the time in seconds needed to operate the design, the better
2: Reliability The design should correctly dispense the target amount and function for longer than the lifespan of the moisturizer.	1: Each dispensation cycle should dispense a volume of product within $\pm 18\%$ of the targeted amount.	1: The closer the mechanism gets by percentage to dispensing the desired amount, the better
	2: The design should be able to withstand a minimum of n cycles, where $n = (\text{volume of moisturizer in the container}) / (\text{volume of moisturizer dispensed each cycle})$	
	3: The design should remain functional for a minimum of 3 years.	

3: Mobility The design should be freestanding and easy for users to move around.	1: Zero permanent installation attachments are required to operate the design.	
	2: The mass of the design when containing none of product shall not exceed 2.5kg	2: The lighter the mass of the design in kg, the better

4. Recommended Design - Transformable Pump/Jar + Crater

This design consists of three main components: a bottle with a pump mechanism that can be twisted open in the middle, a concave bottom, and an additional lid attached to the bottom of the container [Figure-3]. This design initially functions as a conventional pump dispenser with a concave bottom. Users can convert the design to a jar by twisting off sections 1 and 3 and attaching section 3 as a lid for section 2, forming a jar [Figure-3].



Figure-3. The three main components of the design. 1) The top half of the container with the pump. 2) The bottom half of the container with the concave bottom. 3) The attached lid that converts the bottom half to be a jar.

This design addresses the opportunity of residue by allowing users to turn the design into a jar, allowing users' fingers to reach any residue in the bottle. The concave shape allows lotion to be accumulated at one point, enabling the pump dispenser to extract more lotion before the user removes the pump top.

4.1. Key Design Decisions

The key design decisions are inspired by the following diverging tools: reverse brainstorming, SCAMPER & morph, and biomimicry. These various tools enabled exploration of the pump dispenser's features from different perspectives (the container, dispensing mechanism, etc) and the brainstorming of unique designs that fulfill the design requirements [Figure-4].

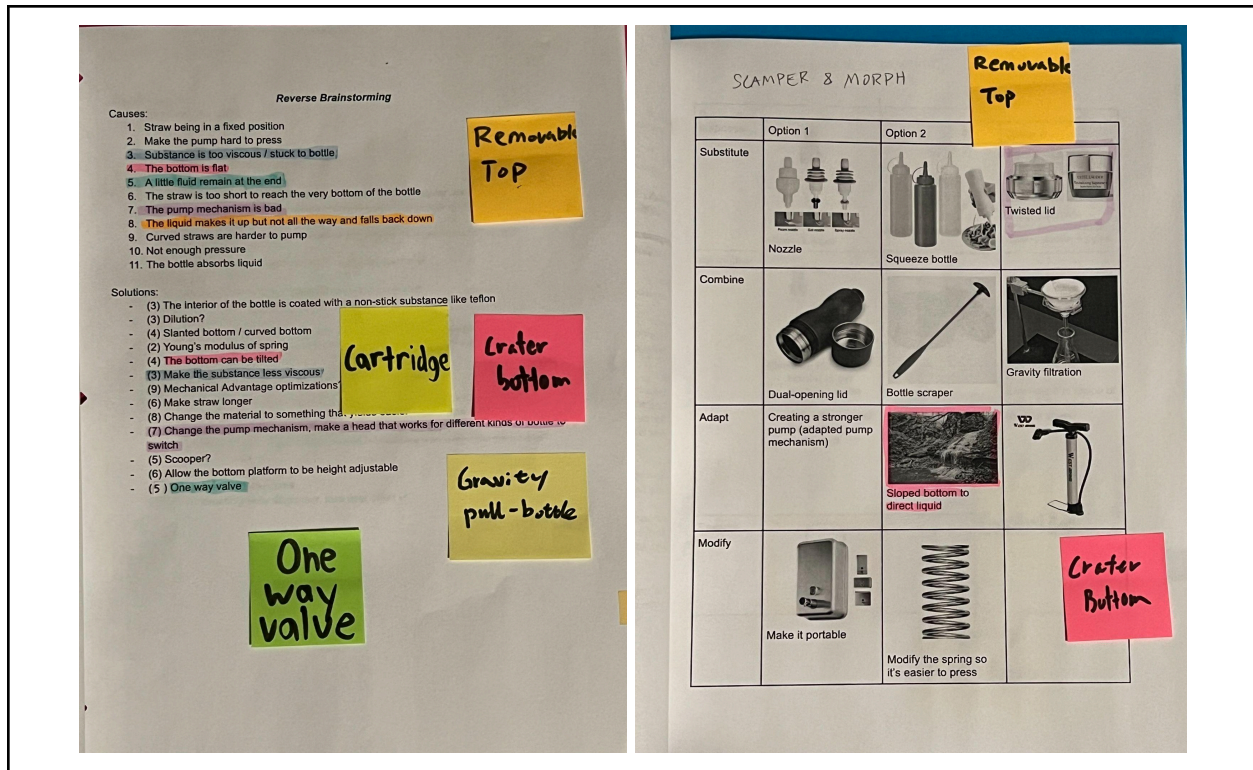


Figure-4. Images illustrating the usage of reverse brainstorming and SCAMPER & morph in the diverging process. The text highlighted purple indicates the ideas that inspired the *Transformable Pump/Jar*, which was referred to as *Removable Top* previously.

See Appendix-G for a [video link](#), presenting visuals of the prototype of the final recommended design in alignment with the key design decisions below.

4.1.1. Jar and Pump Combination

This design can be separated in the middle of the body. Users can unscrew the top half pump mechanism and use a removable lid at the bottom of the bottle as the new lid of the product [Figure-5].



Figure-5. The transition from pump bottle to jar mechanism.

Reverse brainstorming inspired us to think about the jar mechanism as a substitute for the pump mechanism to reduce residue, while SCAMPER inspired us to think about combining the pump and jar mechanisms together in the design.

A study on the residue left by different dispensation methods found that pump dispensers tend to leave a residue of 11.69%, while jar mechanisms only leave a residue of 0.80% [4], [Figure-6].

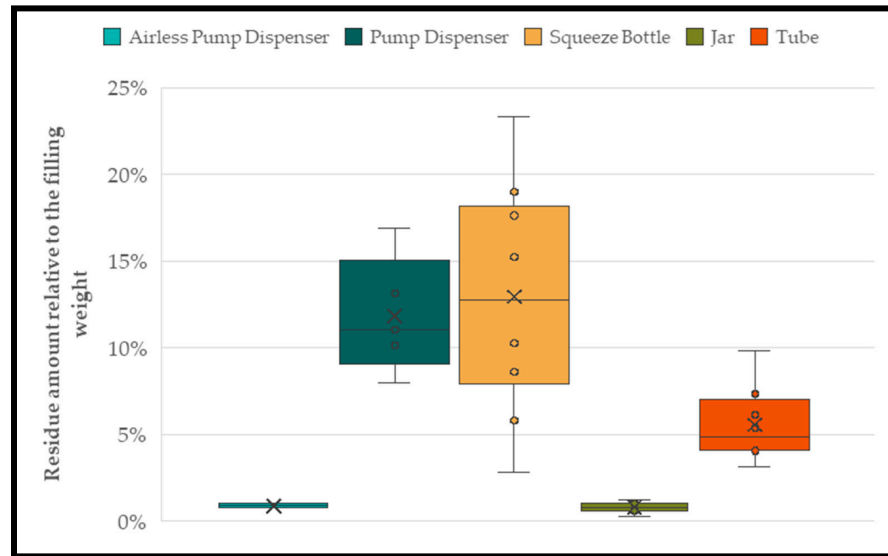


Figure-6. Data on residue left by various body lotion containers.

When residue is more than 11.69%, users can use the pump form and spend less time and force operating the design (6.2) and when it is less, they can use the jar mechanism to extract more residue.

4.1.2. Add-on of the crater

The *Crater* is a concave bottom that allows for the accumulation of lotion to help the pump better extract said lotion [Figure-7]. The crater is inspired by biomimicry, specifically the Kerio Crater, a volcanic crater that has accumulated enough liquid through its concave bottom to form a lake [12]. When converging, a Pugh chart analysis inspired the combining of the crater and pump/jar design. It was noted that the *Crater* could be incorporated into other designs, and its inclusion would be beneficial to the jar/pump design, leading to the two designs being combined. This will be explained further in 6.3.



Figure-7. Crater shape at the bottom of the Pump-Jar prototype.

4.1.3. Additional Attached Lid at the Bottom

The extra lid at the bottom above the crater allows users to close the jar easily and securely, without needing to transform the design back to its pump form [Figure-8]. The jar form allows for an additional 8.2% of the moisturizer to be extracted which will last users for an additional 3.1 days on average [Appendix-F]. For this duration of time, the user can easily change the pump to a jar using the extra lid attached at the bottom of the bottle, which will reduce the space needed to store the bottle and the effort needed to put the lid back.



Figure-8. Removable bottom lid of Transformable Pump/Jar prototype.

5. Other Designs Considered

Multiple designs were brainstormed during the diverging phase of the project. Below are descriptions of the 4 other designs that were focused on during the comparison and converging process [Figure-9].

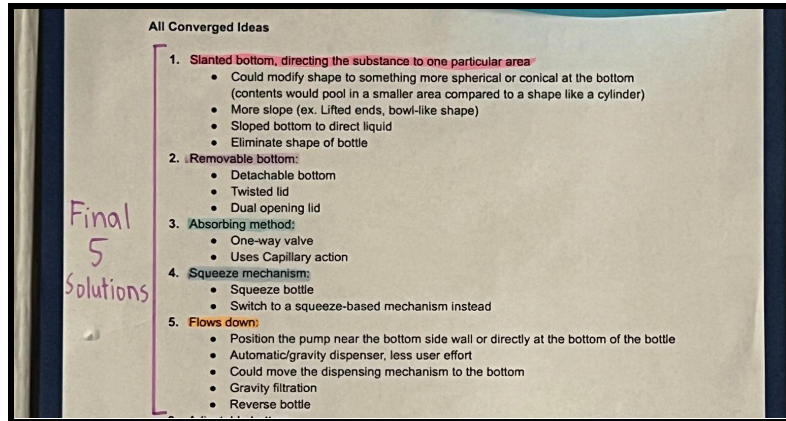


Figure-9. The finalized list of 5 designs that were chosen to be investigated during comparison and convergence.

5.1. Upside-Down Bottle Holder

The upside-down bottle concept flips the pump dispenser upside down and removes the straw, allowing moisturizer to naturally flow downwards towards the dispenser through gravity [Figure-10]. This design will allow users to access residue stuck at the bottom of moisturizer containers, enabling them to use an extra 4.54% of the product while requiring less force to operate (6.2).

This design was inspired by challenging assumptions and SCAMPER; they inspired thought about placing the pump at the bottom and taking advantage of gravity to pull moisturizer downwards respectively. The biggest con of this design is it does not accurately dispense the amount desired, having a variance of 38% which is higher than EC-2 [13].

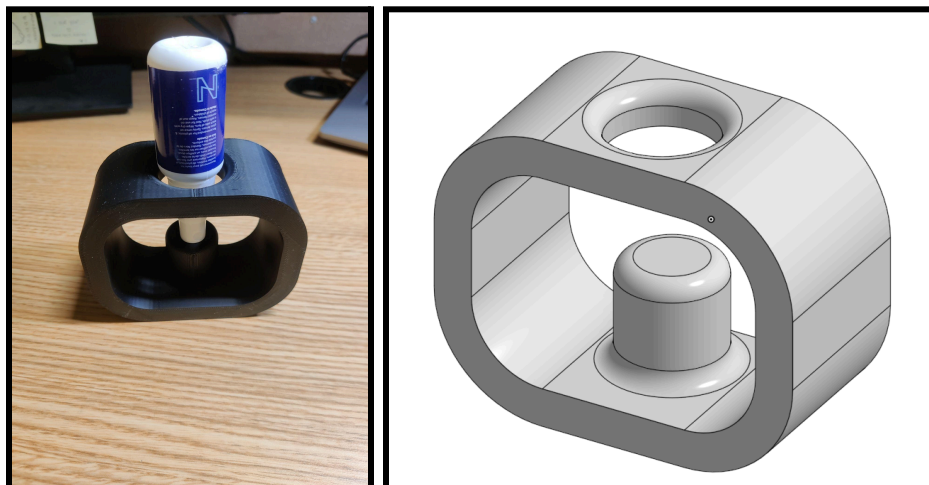


Figure-10. Upside-down bottle holder prototype.

5.2. Modular Cartridge System

This design consists of a plastic bag filled with the moisturizer being sealed inside a shell, with a pump attached to the top of the shell [Figure-11]. When the pump fails to work, users can open the shell and squeeze the residue from the bag, and put it in a new bag. This cartridge system was inspired by milk bags and containers, with the outer shell of this design being analogous to the container of a milk bag. The biggest con of this design is it requires 7 more newtons of force and up to 3.1 seconds longer to operate while retrieving 4.73% less residual than the recommended design (6.2). In addition, it has been found that plastic containers can chemically react with “items containing oils or chemicals, such as cleaning supplies or personal care products,” which can be quite dangerous for users [14].

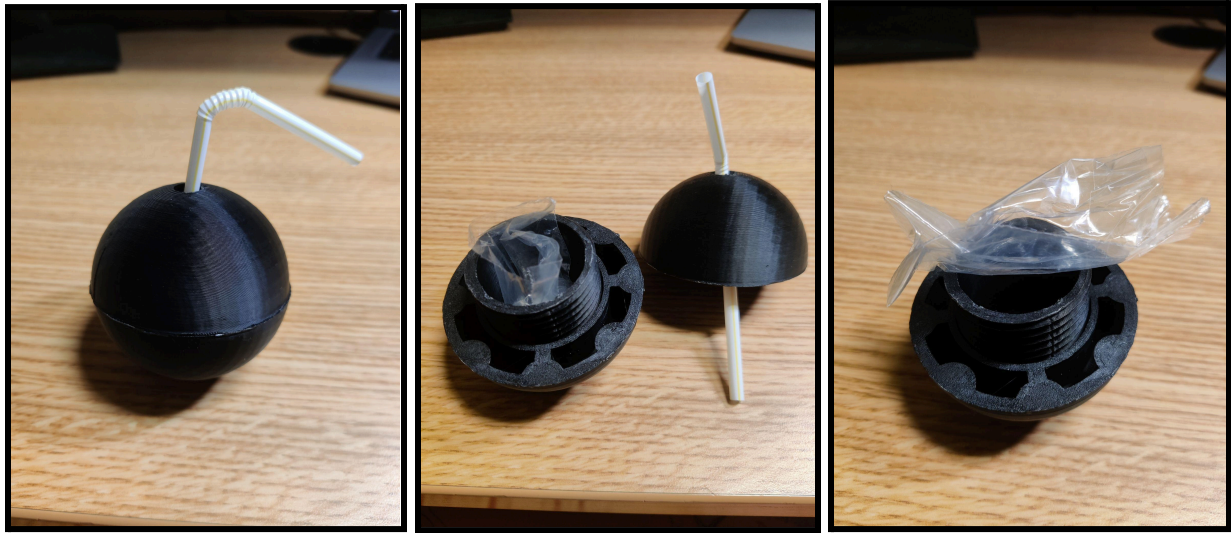


Figure-11. Stages of use of the modular cartridge system prototype.

5.3. Crater

This design allows residue to accumulate and be extracted from a single position through a concave bottom as seen in Figure-12. Many crater shapes were tested [Appendix-D], and the ideal shape that directs the maximum amount of fluid to its center can be seen in Figure-13. The biggest con of the crater is its inability to address the residue remaining on the side of the bottle; the crater can only extract 91.03% of the moisturizer, less than 3 other designs (6.2).



Figure-12. Crater base case shape with an object inside to demonstrate its depth.

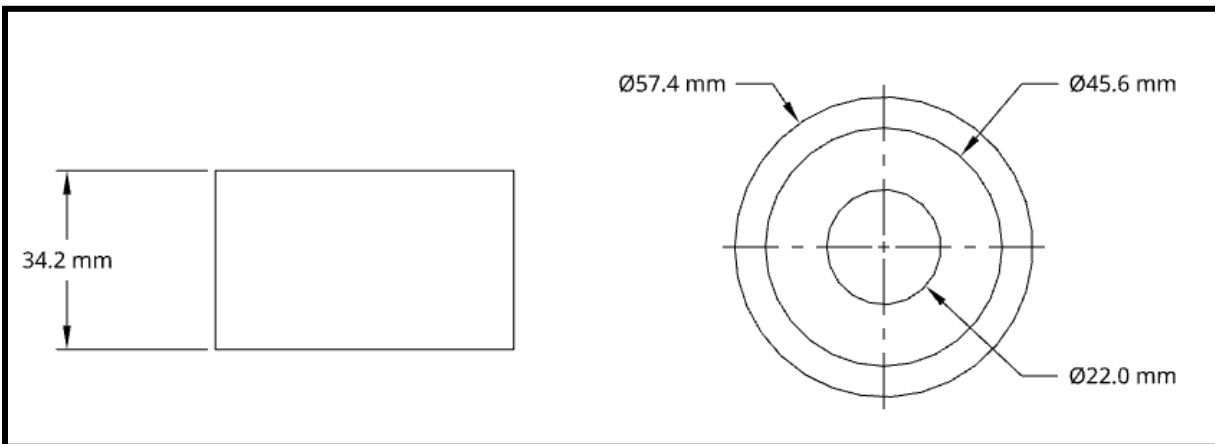


Figure-13. Dimensions of an ideal crater base. Note that this can be scaled to meet any bottle diameter requirement.

5.4. One-Way Valve in the Pump Straw

A one-way valve allows the substance to not fall back down after being extracted into the straw through a ball that blocks the way [Figure-14]. The one-way valve was inspired by biomimicry, specifically capillary action which allows plants to transfer water upwards despite gravity's influence. The biggest con with the one-way valve is that it only prevents backflow, failing to address any residue that the straw itself is unable to reach, such as any stuck on the side of the bottle. This causes it to perform poorly in comparison to other designs, only extracting an additional 1.50% of the moisturizer [15].

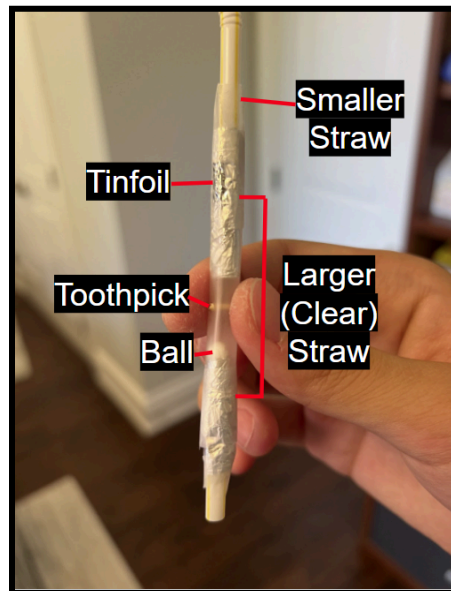


Figure-14. One-way Valve in a straw that prevents moisturizer from going back down to the bottom.

6. Comparison and Evaluation of Candidate Designs

Data was collected through a combination of primary and secondary research. Testing methods and values can be found in the appendices.

6.1 Pairwise comparison

A pairwise comparison was conducted amongst the potential designs. Crater ended with the lowest score, signifying that it is the best design. To validate this result, more evaluations were conducted using measurement matrices and Pugh charts.

Table-2: Pairwise comparison of potential design solutions

Design Name	Crater	Upside Down	Transformable Pump/Jar	Cartridge	One-way Valve	Sum
Crater	—	0	0	0	0	0
Upside Down	1	—	1	0	0	2
Removable Lid	1	0	—	1	0	2
Cartridge	1	1	0	—	0	2
One-way Valve	1	1	1	1	—	4

6.2 Measurement Matrix

For the Jar form of the *Transformable Pump/Jar* and the Bag Form of the *Cartridge*, the accuracy of the dispenser has been recorded as N/A, as the dispensation mechanisms are not mechanically based — they rely on users accessing the product with their fingers.

Table-3: Measurement matrix of potential design solutions

Evaluation Criteria	Reference – Conventional Pump Bottle	Crater	Upside Down	Transformable Pump/Jar		Cartridge		One-way Valve
				Pump Form	Jar Form	Pump Form	Bag Form	
EC-1	88.31% [4]	91.03% [4]	92.85% [4]	99.20% [4]		94.47% [4]		89.81% [15]
EC-2	22N [16]	22N [16]	<22N [16]	22N [16]	29N [17]	22N [16]	29N [17]	22N [16]
EC-3	0.9s [18]	0.9s [18]	0.9s [18]	0.9s [18]	Up to 4s [Table E1]	0.9s [18]	Up to 4s [Table E1]	0.9s [18]
EC-4	Up to 15% [13]	Up to 15% [13]	Up to 38% [13]	Up to 15% [13]	N/A	Up to 15% [13]	N/A	Up to 15% [13]
EC-5	0.08kg	0.12kg	0.17kg	0.08kg	0.06kg	0.07kg	0.002kg	0.085kg

6.3 Pugh Charts

Round 1

Due to the transformable pump/jar and cartridge having two modes of dispensation, their modes are being evaluated independently for all criteria aside from volume extracted. This is because the maximum volume extracted will always be attained while the designs are in their non-pump forms.

The solutions that are comparatively the worst are the *conventional pump mechanism* and the *one-way valve*. They have a “-” in volume extracted, which has been highlighted by stakeholders as one of the most important criteria, and their only + is not unique as the majority of other designs are also better than the reference at meeting the mass criteria. Therefore, the 4 other designs have been chosen to be evaluated during round 2.

Table-4: Pugh Chart Round 1 - All prototypes and a conventional moisturizer bottle

Evaluation Criteria	Reference – Crater	Conventional Pump Mechanism	Upside Down	Transformable Pump/Jar		Cartridge		One-way Valve
				Pump Form	Jar Form	Pump Form	Bag Form	
EC-1	Same	-	+	+		+		-
EC-2	Same	Same	+	Same	-	Same	-	Same
EC-3	Same	Same	Same	Same	-	Same	-	Same
EC-4	Same	Same	-	Same	+	Same	+	-
EC-5	Same	+	-	+	+	+	+	+
Analysis	5 Same	3 Same 1+ 1 -	1 Same 2 + 2 -	3 Same 4 + 2 -		3 same 4 + 2 -		2 Same 1 + 2 -

Round 2

Analysis shows that *crater* is the worst remaining solution. It is worse than the reference at meeting the key criteria of volume extracted unlike other solutions, while having benefits that are not unique. Thus, the *upside down bottle holder*, *transformable jar/pump*, and *cartridge* have been chosen to be evaluated during round 3.

Table-5: Pugh Chart Round 2

Evaluation Criteria	Reference – Upside Down	Crater	Transformable Pump/Jar		Cartridge	
			Pump Form	Jar Form	Pump Form	Bag Form
EC-1	Same	-	+		+	
EC-2	Same	-	-	-	-	-
EC-3	Same	Same	Same	-	Same	-
EC-4	Same	+	+	+	+	+
EC-5	Same	+	+	+	+	+
Analysis	5 Same	1 Same 2 + 2 -	1 Same 5 + 3 -		1 same 5 + 3 -	

Round 3

When comparing the transformable pump/jar and the cartridge, the pump forms were compared, and the non-pump forms were compared.

Analysis of round 3 shows that the *Transformable Pump/Jar* is the best solution remaining.

The *Cartridge System* is better than the *Transformable Pump/Jar* at meeting the mass criteria, however, it is worse than the Pump/Jar at meeting the volume criteria. Since volume extracted is a key criteria, the *Transformable Pump/Jar* has been concluded to be the more suitable solution.

The *Upside Down Bottle Holder* is better at meeting 3 criteria while being worse at meeting 5 criteria. Of these 5 criteria, one is the key criteria of volume extracted. Since the *Upside Down Bottle Holder* is worse at fulfilling the key criteria of volume, while also fulfilling more criteria worse than better, it has been concluded to be a less suitable design than the *Transformable Pump/Jar*.

Therefore, the *Transformable Pump/Jar* has been found to be the best design according to the Pugh charts.

Table-6: Pugh Chart Round 3

Evaluation Criteria	Reference – Transformable Pump/Jar		Cartridge		Upside Down	
	Pump Form	Jar Form	Pump Form	Bag Form	Compared to Pump	Compared To Jar
EC-1	Same		-		-	
EC-2	Same	Same	Same	Same	+	+
EC-3	Same	Same	Same	Same	Same	+
EC-4	Same	Same	Same	Same	-	-
EC-5	Same	Same	+	+	-	-
Analysis	9 Same		6 same 2 + 1 -		1 Same 3 + 5 -	

Round 4

One feature noted about the crater was its ability to be easily combined with other designs. Thus, an evaluation was conducted into whether its inclusion with the most suitable design, the *Transformable Pump/Jar*, was beneficial or not.

The crater enables more residue to be extracted before a transformation to jar form is needed. This leads to the amount of force required and time spent operating the design being, on average, less across the product's entire lifespan. Though the crater does increase the weight of the design, the weight of the combined design still falls within the weight requirement of 2.5kg, defined by EC-5. Thus, the benefits of adding the crater have been found to outweigh the cons, indicating that its inclusion is overall beneficial to the design.

Table-7: Pugh Chart Round 4 - Combination of the Crater and Transformable Pump/Jar Design

Evaluation Criteria	Reference – Transformable Pump/Jar on its own		Crater on its own		Transformable Pump/Jar + Crater	
	Pump Form	Jar Form	Compared to Pump	Compared to Jar	Pump	Jar
EC-1	Same		-		Same	
EC-2	Same	Same	Same	+	Same	+
EC-3	Same	Same	Same	+	Same	+
EC-4	Same	Same	Same	-	Same	Same
EC-5	Same	Same	-	-	-	-
Analysis	9 Same		3 same 2 + 4 -		5 Same 2 + 2 -	

Therefore, the Transformable Pump/Jar + Crater design is justified to be the recommended solution to the presented opportunity.

7. Conclusion

This report recommends a Transformable Pump/Jar + Crater design, making use of an adaptable bottle mechanism shape and a concave crater bottom to reduce residue waste at the end of a pump bottle's lifecycle. It addresses the need of cost-conscious first-year students to maximize the amount of fluid extracted from moisturizer pump bottles while avoiding permanent instillations. Testing of this solution demonstrated that it can perform as a jar to facilitate user extraction of near-empty containers, and that the crater bottom successfully directs fluid to its center. In the future, a wider product scope, such as serums or shampoos, can be tested to strengthen the features of the current design to be multifaceted.

8. Appendices

Appendix A - Original Design Brief

*******Note: The Original Design Brief is highlighted in GRAY to differentiate any titles or citations seen.*******

Design Brief — The Inconvenient Dispensing Mechanism of Household Body Care Products

Word count: 1772

1. Introduction: The Problem with Current Dispensers

This design brief outlines the inconvenient limitations of current dispenser designs in household body-care products, and develops a series of guidelines to frame a design space for a solution.

Fluid body care products are a household staple for millions, and often use classic pumps for dispensing. However, the limited reach of the current pump design makes it difficult to extract product as contents run low. This serves as a major source of frustration for consumers, as significantly more effort is needed to perform what once was a simple task. This issue has proved to be widespread among consumers—a survey reveals the overwhelming majority (approx.

91.55%) of respondents have experienced such difficulties (See Appendix A2). This report delves into current inconvenient designs and establishes clear engineering objectives focused on enhancing user experience.

1.1 Stakeholders and Their Values

On the highest level, the key stakeholders for this opportunity are typical household consumers. However, those with mobility limitations also need to be considered.

Investigations reveal that the increase in force and number of pumps required to extract the last bit of product leads to unnecessary fatigue and even physical risks. An interview with an individual suffering from cerebral palsy provided a first-hand account of an inability to dispense a product due to uncoordinated hand movements, forcing them to manually pour out the contents of the container to retrieve the desired amount. (See Appendix A8 for full statement). This account reflects the struggles of many individuals with disabilities and reiterates the need for a design supporting ease-of-use and accessibility.

In addition to increased effort, the extra time required to extract the residue causes mental strain. A survey of 140 individuals revealed that 36.4% of respondents feel mildly infuriated using the dispenser at this stage, with a combined 12.9% (See Appendix A4 for the graph) expressing major agitation. These first-hand accounts concluded that the dispenser requires improvements in order to meet user needs.



Figure 1: Lotion bottles with residue leftover that the pump mechanism cannot reach. The only way for the consumer to access the rest of the contents was to remove the pump and scrape the inner contents out. [11]

1.2 The Definition of Convenience

Within the context of this study, “convenience” is defined as a composite parameter characterized by accessibility, reliability, consistency, and efficiency, with a focus on minimizing the time and effort needed to operate the mechanism.

2. Reference Designs: Alternative Designs with Limitations

This section highlights some existing alternative designs that meet some stakeholder needs at the expense of others. These reference designs guided the development of many of the

objectives—understanding their limitations will help frame the boundaries of the design space defined in section 3.



Figure 2: Wall-mounted soap dispenser often seen in public washrooms. Note that the location where soap is dispensed is at the *bottom* of the container, compared to typical freestanding designs where the soap is dispensed from the top. [13]

2.1 Wall-Mounted Soap Dispensers

Wall-mounted soap dispensers are commonly used in public washrooms to dispense soap efficiently, leveraging gravity to guide fluid to the bottom opening. However, this design is rarely seen in household settings as it does not meet values of mobility and replaceability. Users value the convenience of buying body-care products that can be moved easily throughout the home, requiring zero installations.



Figure 3: Silicon pipe pump mechanism seen in some cosmetic products. Note that the flexible tube allows for more residue to be reached, but costs an extra amount of user effort to maneuver the pipe to reach the residue. [14]

2.2 Bottles With Flexible Hoses

Designs with flexible silicone pipes are another solution to the issue of residue but require additional force to operate, impeding on efficiency and convenience. In a cosmetic press pump, the mobile hose maximizes access, however fluids with a higher viscosity will require extra force and effort to dispense through the tube's curvature. In addition to this, the benefits of a flexible hose can only take effect if the user manually twists and moves the tube, requiring time and effort to retrieve the last few drops of fluid.

3. Needs, Goals, and Objectives: A guideline to meeting stakeholder values

The following section outlines a set of a need, goals, and objectives to better guide the development of future designs. Each was crafted with careful consideration of the stakeholder values established in section 1.1 and the shortcomings of the reference designs described in section 2.

3.1 Design Need and Rationale

We need a dispensing mechanism that will conveniently dispense all of the container's contents with minimal user effort no matter the level of fluid in the container.

This need statement effectively captures the stakeholder values from section 1 by explicitly targeting convenience, the level of user effort, and the effectiveness of the mechanism itself. The following sections will break this statement down into 4 specific goals and corresponding objectives to further frame the design space.

3.2 Creating the Design Space: Goals and Corresponding Objectives

To address this need, the following goals were established to guide the improvement of the dispensing mechanism. Many were developed under the guidance of ISO's standards for ergonomic human-system interactions, as convenience is inherently a product of the quality of user experience. According to their recommendations, a design where diverse users can overcome any physical or psychological barriers and access the system to accomplish the desired task is more ergonomic and hence more convenient [9].

The following goals were also formulated using relevant DfX principles, including Design for Ergonomics [5][6][9][10], Design for User Experience [7], Design for Reliability [4], and Design for Accessibility [3].

3.2.1 Goal 1: Functionality

The design should immediately and effectively dispense fluid.

This goal targets the most fundamental design function—effectively dispensing fluid with no residue. The time specification accommodates the expected immediate nature of fluid dispensing. One should not have to wait for fluid to be dispensed, and so this goal directly addresses the stakeholder value of quickness and ensures quality functionality in the design.

Objective 1

The system should execute the intended task within 1 second of operation.

This objective is derived from the principles of User Experience design, stating the system should output results within 1 second of user input to retain user engagement [7]. While this recommendation was developed for user interface design, it also provides a useful limit for mechanical systems looking to prioritize immediacy of output.

Objective 2

The dispensing mechanism should be able to extract $\geq 98\%$ of the product.

This objective targets the residue left behind by classic pump mechanisms. In order to extract all product, users must open the bottle and pour the rest, causing inconvenience. New designs should mitigate this issue by ensuring that nearly all of the product can be extracted without additional user effort.

3.2.2 Goal 2: Reliability

The design should be reliable.

Reliability requires consistent performance throughout the product's lifetime to target the user's sense of trust in the design. ISO's ergonomic standards recommend that trustworthy designs take into account the *entire* user experience [4], which includes the entire duration of the design's life. Convenience should be consistent at every stage of use—the quality of output should not deteriorate with time.

Objective 1

Each dispense cycle should dispense a volume of product within $\pm 10\%$ of the target dispense amount.

This objective ensures that the design has consistent performance throughout its lifetime. By maintaining a consistent volume of fluid dispensed each cycle, the level of effort required to operate the design will also remain consistent.

Objective 2

The design should be able to withstand a minimum of n cycles, where:

$$n = \frac{\text{Volume of fluid in container}}{\text{Volume of fluid dispensed each cycle}}$$

This objective ensures that the design as a whole can reliably perform for the entire duration of the product's life, which is dependent on the volume of fluid in the container. gives the theoretical number of dispenses it would take to extract all fluid from the container. n

3.2.3 Goal 3: Accessibility

The design should be operable by those with mobility limitations.

This goal targets convenience for people of different physical abilities. ISO standards state that for a system to be ergonomic, diverse users should be able to overcome any physical or psychological barriers to accomplish the task [9]. When content runs low, the act of continuously pumping product becomes fatiguing as more pumps are generally needed, especially for those with physical limitations. The following objectives serve to mitigate this.

Objective 1

No more than 30 N of force should be required to operate the product per dispensing cycle.

ISO's recommendations for designing for accessibility suggests that the minimum comfortable pushing strength for diverse users (with one finger) is 40-60 N for male adults, and 30-60 N for female adults [10]. Since the use of a finger strongly reflects the action used to apply a force on a dispenser's actuator, taking the lowest boundary (30 N) ensures that the product is accessible to the widest range of individuals [3].

Objective 2

The surface area of the actuator should be at minimum of 1 cm².

This property was derived from ISO's recommendation that any buttons requiring physical input have a minimum diameter of 1 cm for easy access [1]. As non-circular buttons also lie within this design space, the standard was extrapolated to a surface area of 3 cm², allowing for a range of physical input designs while maintaining a standard of comfort for every individual.

3.2.4 Goal 4: Mobility

The design should be freestanding and mobile.

This goal targets a major stakeholder value pertaining to a household product, highlighted in section 2 through the reference design of wall-mounted soap dispensers. The design must be mobile and require minimal effort to move.

Objective 1

Zero permanent installation attachments are required to operate.

No permanent installation attachments means the design requires no extra wall-mounting or other sort of permanent fixation to be used, thus making it free-standing and moveable.

Objective 2

The design's weight when empty should not exceed more than 10% of the design's weight when filled with fluid.

The design should be lightweight in order to minimize user effort regarding lifting and relocation. Secondary research yields that even the heaviest type of household soap dispensers on the market—glass refillable dispensers—do not exceed an empty weight of more than 10% of the weight when filled (See Appendix C).

4. More Reference Designs: Promising Design Features

Further exploration of reference designs yielded two design features with potential to satisfy the goals outlined in section 3. One makes use of a partial vacuum, and the other utilizes the container's geometry to extract more lotion, expanding the design space beyond the conventional pump dispensing system. However, there are limitations associated with both designs as a whole. See Appendix B for detailed descriptions.

5. Conclusion

This design brief defines an opportunity to improve the convenience of household fluid dispensing mechanisms by analyzing stakeholder values and outlining a set of specific needs, goals, and objectives to develop a design space. It is imperative that any engineer looking to work with this opportunity maintains a focus on ergonomics, accessibility, and reliability to effectively answer user needs and improve the convenience of operating a fluid dispenser.

6. Source Extracts

[9]

5.3.7 Approachability

ISO 9241-110:2020, 5.7 goes beyond this avoidance of barriers and introduces the principle of “user engagement”, which it describes thus: “The interactive system captures the users' attention and motivates the users to continue to use the interactive system.”

NOTE ISO/IEC Guide 71:2014, 6.2.4 includes a goal it calls “approachability”, which it describes thus: “A system is approachable if diverse users can overcome any physical or psychological barriers and physically or remotely access it to accomplish the task.”

ISO 9241-110 provides general design recommendations related to user engagement, including guidance on:

- motivating the user;
- trustworthiness;
- increasing user involvement.

[3]

1236

- Using absolute reference controls (e.g., pointers) rather than relative controls (e.g., pushbuttons to increase/decrease, or round, unmarked knobs)
- Using moving pointers with stationary scales
- Providing multisensory indications of control status (e.g., in addition to a status light indicating “on,” provide an intermittent audible tone and/or tactilely discernible vibration)
- Using direct keypad input
- Providing speech output to read or confirm the setting

See Sections O-3, O-4, and O-5 for design options covering visual displays. See Figures 12 and 13.

9.2.5 I-5 Maximize the Number of People Who Can Physically Operate Controls and Other Input Mechanisms (Operate)

Problem Controls (or other input mechanisms) may be difficult or impossible for those with physical disabilities to operate effectively.

Example: People with severe weakness may be unable to operate controls at all or may have great difficulty performing constant, uninterrupted input. People with only one arm or without arms (but utilizing assistive devices such as headsticks or mouthsticks) may not be able to activate multiple controls or keys at the same time. People with artificial hands or reaching aids may have difficulty grasping small knobs or operating knobs or switches which require much force. People with poor

DESIGN FOR INDIVIDUAL DIFFERENCES

coordination or impaired muscular control have slower or irregular reaction times, making time-dependent input unreliable. People lacking fine movement control may be unable to operate controls requiring accuracy (e.g., a mouse or joystick) or twisting or complex motions. People with limited movement control (including tremor, poor coordination, or those using headsticks or mouthsticks) can inadvertently bump extra controls on their way to a nearby desired control.

Design Options and Ideas to Consider

- Minimizing the need for strength by minimizing force required as much as possible or by providing adjustable force on mechanical controls
- If stiff resistance is provided to prevent accidental activation, it could drop off after activation. Other non-strength-related safety interlocks could also be considered.
- Spacing the controls to provide a guard space between controls, thus also leaving room for adaptations such as attaching levers to hard-to-turn knobs or room to replace knobs with larger, easier-to-turn knobs or cranks
- Minimizing or providing alternatives to performing constant, uninterrupted actions (e.g., button locks or push on-push off buttons to eliminate the need to press and hold some buttons continuously)
- Where simultaneous actions are required (e.g., pressing shift or control key while typing another key), providing an alternative method to achieve a result that does not

[10]

4.3 Strength required for operation

Considerations of population strength characteristics, such as age, sex and physical capabilities and limitations, for operation of a product and controls that require user strength output (e.g. holding, gripping, pushing, sliding) increase accessibility.

The following specific design requirements and recommendations are intended to increase accessibility related to human strength. For the strength of each particular action, see [Clause 5](#).

- Strength for operating a product should be set at the minimum suitable force required for comfort control.

- A product that needs excessive strength for operation should have an additional device to assist, thereby decreasing the required strength.

NOTE Some people with physical limitations of the upper or lower limbs, in particular people with rheumatism or spinal cord injuries, are unable to operate some products.

- When operation of a product requires opening and closing, the actions should require minimal strength to be accessible by the widest range of population (e.g. age, sex, strength limitations), except where a fail-safe mechanism and safety considerations are needed (see [4.9](#)).
- Products, controls and control panels shall have a non-slippery surface finish with purchase for hands or fingers to make operation easier for people with muscle strength limitations.
- When a product uses a touch screen, care should be taken with the appropriate pressing strength.

[1]

8.1.2.5 Buttons

The buttons shall be laid out so that they correspond with the information and content they control and shall be arranged in hierarchical order or in a sequence which will aid recognition and use in accordance with ISO 9241-410:2008/Amd 1:2012.

Buttons should protrude in depth from the surface and have a diameter/width of 10 mm or greater. When a button is pressed, a clear haptic feedback shall be produced. The haptic feedback should not be audible.

Groups of buttons should have tactile markings that aid identification and navigation in accordance with ISO 24503.

NOTE A large number of controls can create complexity and confusion, therefore it is best to keep them to a minimum.

[10]

5.3.2 Strength (hands and fingers)

A product, a control or a control panel that is operated by gripping, grasping, rotating, twisting, pushing or pulling (with fingers), pinching, sliding and touching with hands or fingers should be sufficiently strong for these actions to be performed easily by the widest range of users, taking account of the position of the action (e.g. height), direction of force and posture required to perform the action. The acceptable strength should be based on population strength data that include age differences, sex differences and the effects of physical limitations. Consideration should be devoted to height for wheelchair users, in particular.

A product, a control or a control panel that is operated by the hands and fingers should be designed so that people with limitations of strength or of dexterity of hands or fingers (e.g. because of Parkinson's disease or rheumatism) are able to operate it.

NOTE 1 Data for maximum grip strength for male and female persons of different ages are available from scientific papers^{[23][24][25]} and research project reports (TU Delft,^[21] MEXT^[26]). Male grip strength is in the range of 100 N to 550 N and female grip strength is in the range of 100 N to 350 N, depending on age. They respectively show the highest values at 20 years to 30 years of age, gradually decreasing with age thereafter. The data are useful for designing products that require squeezing with the hands, such as refuelling hoses and pliers. See [B.3.12](#).

NOTE 2 Data for the maximum twisting strength of the wrist have been reported by NITE^[27] and DTI^[18] using circular knobs of 50 mm diameter (NITE) and of 40 mm diameter (DTI) for male and female persons of different ages. The necessary twisting strength is in the range of 3 Nm to 5 Nm for male adults and 2 Nm to 4 Nm for female adults, depending on the grip type and knob type. The strength increases with age from 2 years to 20 years until it reaches a plateau, which remains until 60 years of age. It then decreases. No significant difference was found between inward and outward rotation. The data are useful for designing doorknobs, keys and dials for household appliances, for example. See [B.3.13](#).

NOTE 3 Data for the maximum twisting strength of the upper limbs have been reported by NITE.^[27] The data were sampled from 38 male persons and 42 female persons who were 60 years to 73 years old. The maximum torque strength is about 5 Nm to 8 Nm for male persons and 3 Nm to 5 Nm for female persons. The necessary torque for inward rotation (pronation) is greater than that for outwards rotation (supination). The data are useful for designing large and heavy handles for household appliances, for example. See [B.3.14](#).

NOTE 4 Data for the maximum torque with two hands for jar opening have been reported by DTI^[19] and TU Delft^[21] for male and female persons of different ages. The opening torque ranges from 3 Nm to 10 Nm for male adults and from 2 Nm to 6 Nm for female adults, depending on the jar diameter and the lid type. The typical ageing effect is a rather steep increase up to about 20 years, a plateau in adulthood and a gradual decrease with age thereafter. A significant effect is found for the jar diameter and the lid type, i.e. smooth or knurled. The data are directly useful for designing packages. See [B.3.15](#).

NOTE 5 Data for the maximum pushing strength with a finger have been reported by DTI^[19] and NITE^[27] for male and female persons of different age groups. The DTI data show strength to be in the range of 120 N to 180 N for male adults and 80 N to 130 N for female adults, whereas NITE data show the range as 40 N to 60 N for male adults and 30 N to 60 N for female adults. This difference depends on the measurement method employed: a force plate for DTI and a real push-button for NITE. No large difference is found between forward and downward pushing, but a significant difference exists between the forefinger and the thumb. The data are useful for designing push-buttons of various types used in home appliances. See [B.3.16](#).

4 Principles of human-centred design

4.1 General

This part of ISO 9241 provides a framework for human-centred design. It does not assume any particular design process, nor does it describe all the different activities necessary to ensure effective systems design. It is complementary to existing design methodologies and provides a human-centred perspective that can be integrated into different design and development processes in a way that is appropriate to the particular context. All the human-centred design activities identified in Clause 6 are applicable (to a greater or lesser extent) at any stage in the development of a system.

Whatever the design process and allocation of responsibilities and roles adopted, a human-centred approach should follow the principles listed below (and described in 4.2 to 4.7):

- a) the design is based upon an explicit understanding of users, tasks and environments (see 4.2);
- b) users are involved throughout design and development (see 4.3);
- c) the design is driven and refined by user-centred evaluation (see 4.4);
- d) the process is iterative (see 4.5);
- e) the design addresses the whole user experience (see 4.6);
- f) the design team includes multidisciplinary skills and perspectives (see 4.7).

7. Works Cited

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8. Appendix

Appendix A: Primary Research—Data and Graphs from the Survey (140 responses)

Full survey results:

https://pubref.paperol.cn/clientdown/332887168/332887168_1760042580r2Mc3j.pdf

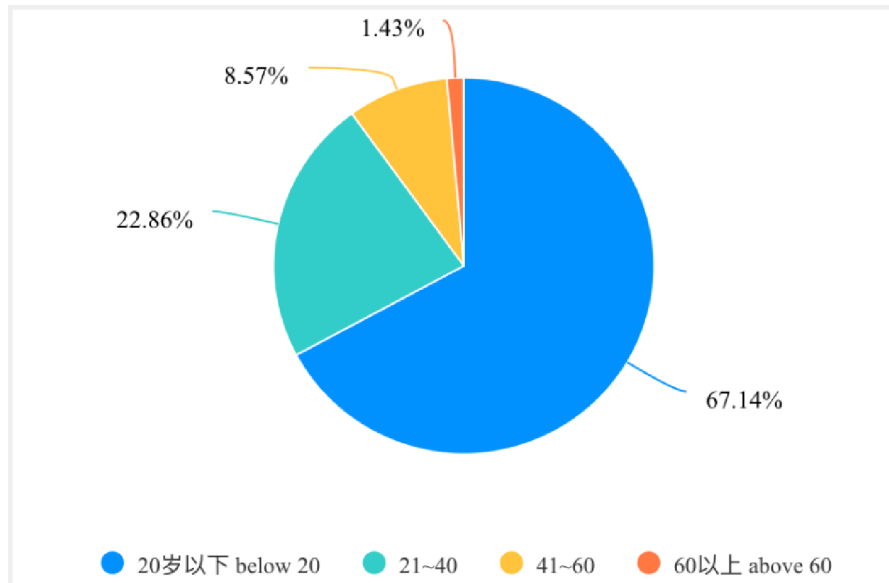


Figure A1: Pie chart showing results from question 1 in the survey which asked for participants' age.

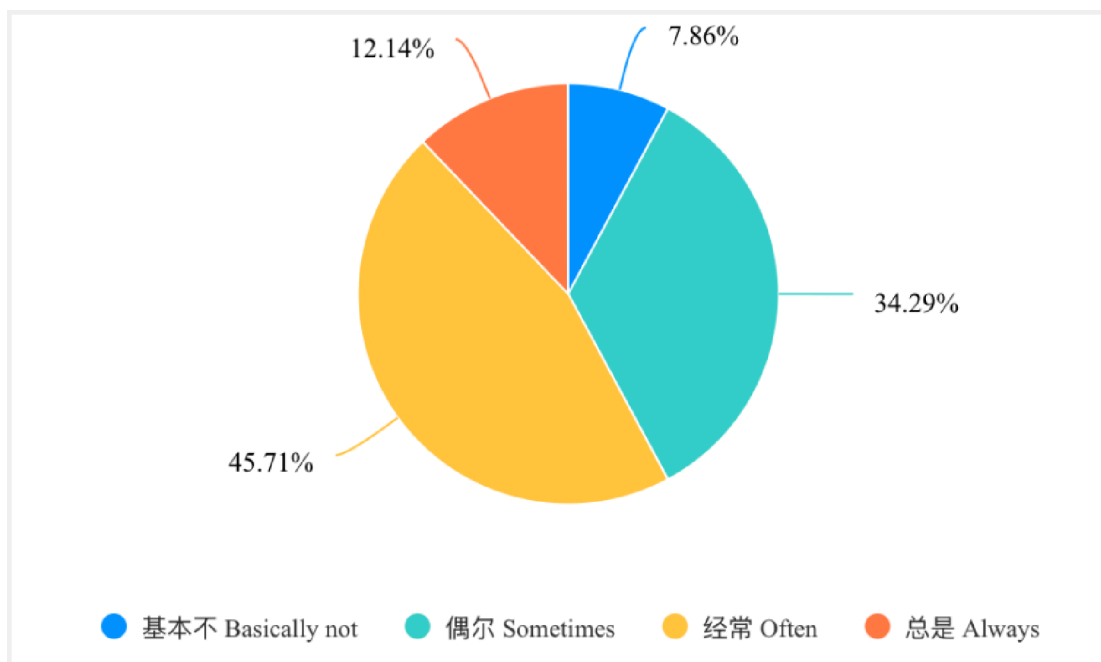


Figure A2: Pie chart showing results from question 2 in the survey which asked the frequency of participants using press-pump bottles in their daily life.

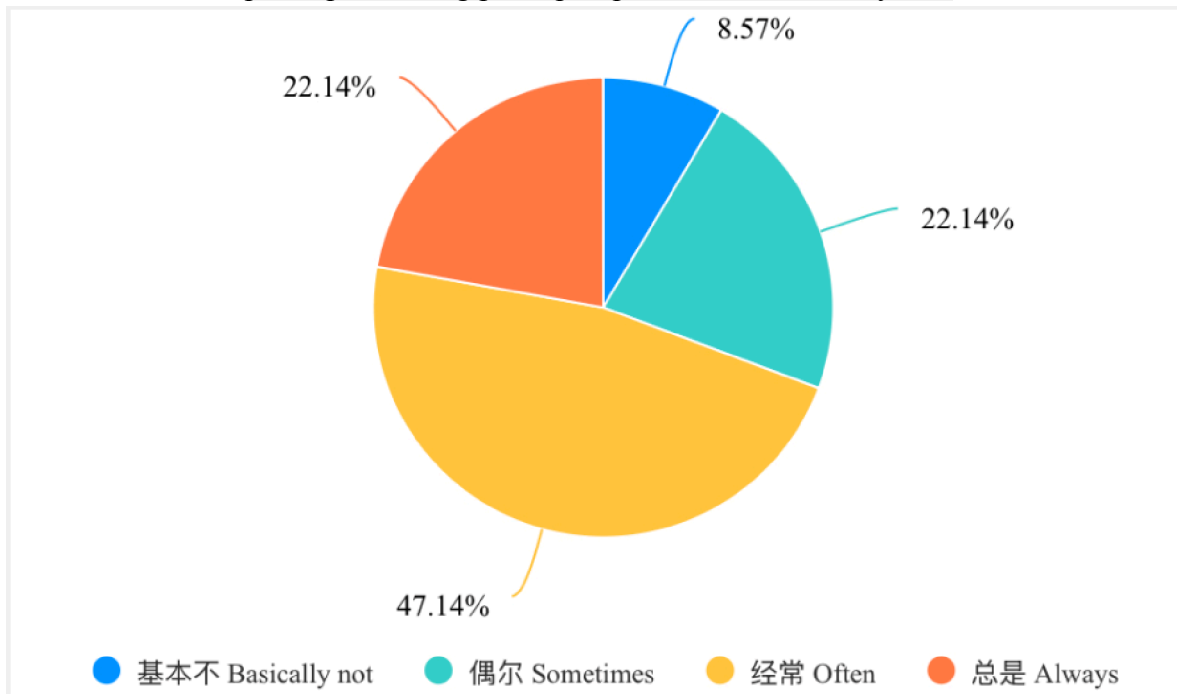


Figure A3: Pie chart showing results from question 3 in the survey which asked the frequency of participants facing the waste problem of press-pump bottles in their daily life.

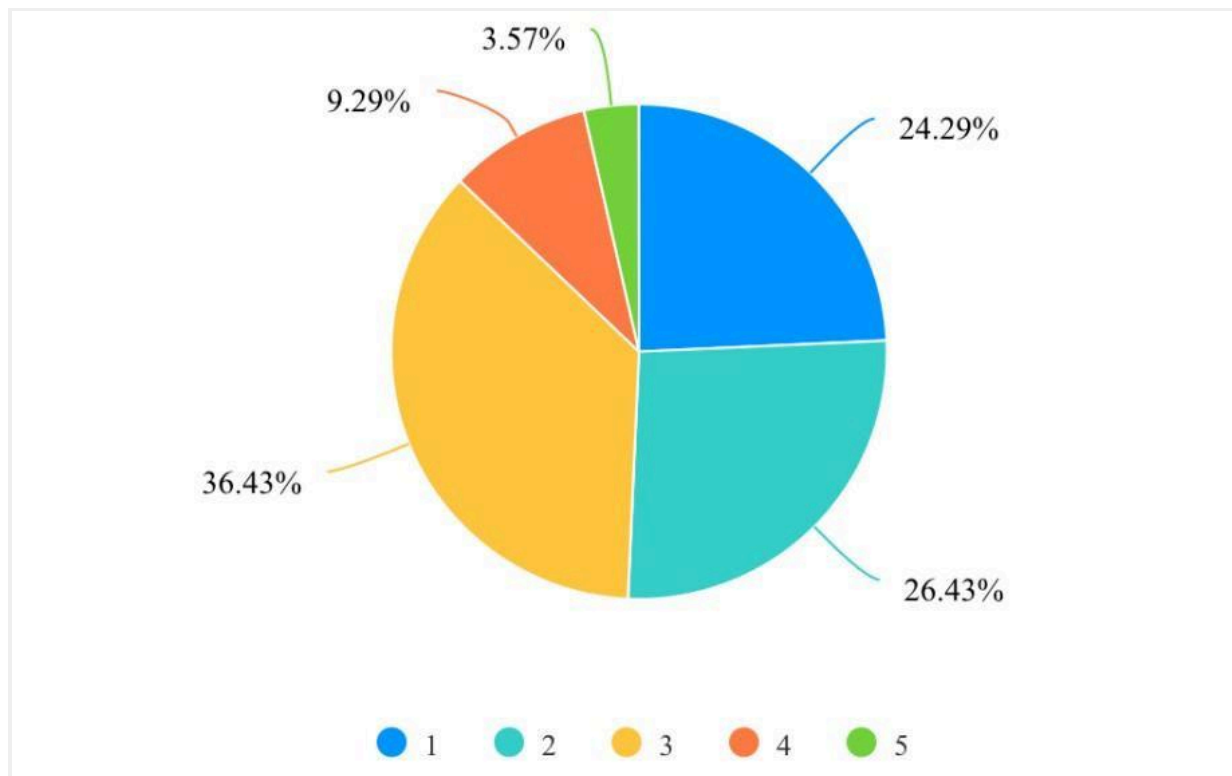


Figure A4: Pie chart showing results from question 4 in the survey which asked the participants'

feelings when they were facing the waste problem (1 is happy, 5 is anger)

总值为：838 平均值为：5.9

Figure A5: The result of question 5 in the survey which asked participants how many pumps they think they need to get the same amount of lotion when the lotion is nearly finished. The number on the left side: 838 is the sum of numbers provided by the participants. The number on the right side: 5.9 is the average value that participants think they need, which is calculated using $838/140$.

总值为：559 平均值为：3.94

Figure A6: The result of question 6 in the survey which asked participants how many times greater is the force when the lotion is nearly finished compared to the force when the lotion is full. The number on the left side: 559 is the sum of numbers provided by the participants. The number on the right side: 3.4 is the average value that participants think they need, which is calculated using $559/140$.

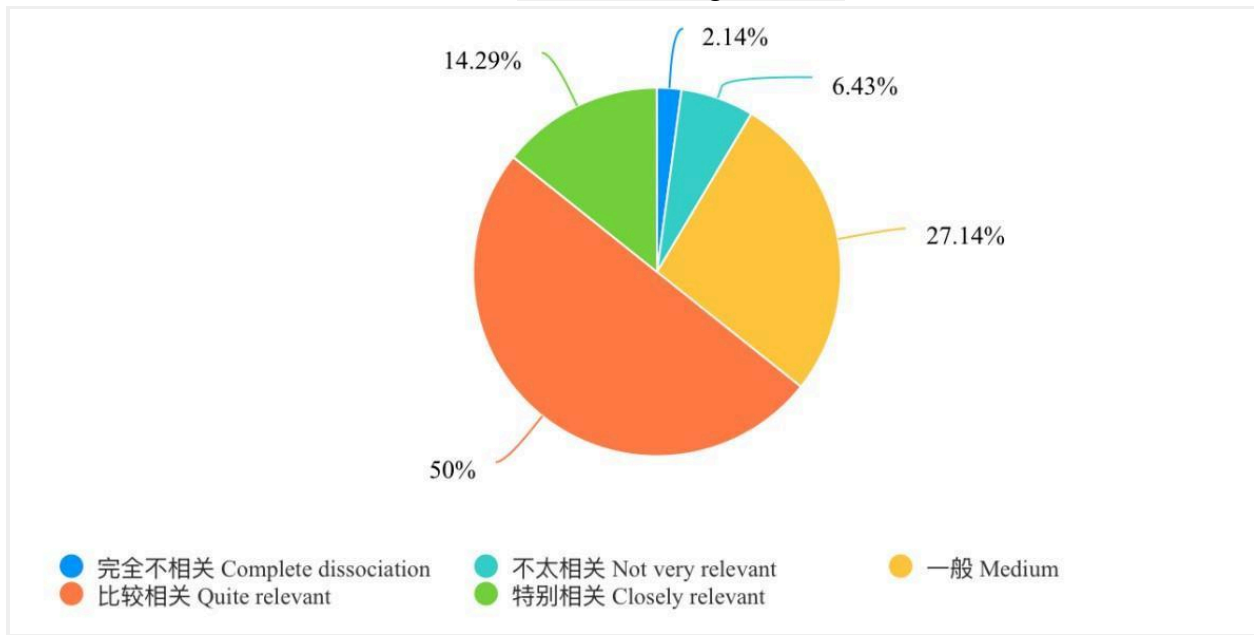


Figure A7: Pie chart showing results from question 7 in the survey which asked the relevance of these kinds of problems (Survey question 3-6) of participants facing the waste problem of press-pump bottles in their daily life.

由于本人身体条件，手部运动不协调，导致无法很好的挤压按压瓶，通常在快用完时我会选择卸下按压器，直接倒出

Figure A8: There are several results collected for question 8 in the survey which asked participants to give any other problems they encountered when cosmetics were almost running out. This open question is not compulsive for every participant to answer. The figure is the result from one participant who had cerebral palsy.

Below is the translation of her result translated through Google Translate:

Due to my physical condition, my hand movements are uncoordinated, making it difficult to properly squeeze or press the bottle. Usually, when the bottle is almost empty, I choose to remove the pump and pour the contents out directly.

Appendix B: Reference Designs with Promising Design Features

Figure B1: Airless pump bottle design featuring a large ergonomic press head. [17]



The interior of the bottle in this reference design is designed to maintain a partial vacuum which can help reduce the lotion waste. During operation, when the user presses the pump, the pressure differential between the inside and outside of the bottle causes the flexible bottom section (with a cross-sectional lifting mechanism) to move upward, which “cleans” all the lotion inside of the bottle.

However, a performance issue has been observed:

in some cases, the pressure differential is insufficient for the bottom section to rise automatically. As a result, users may need to manually press the bottom of the bottle to enable normal operation, adding a level of additional effort.

Figure B2: Funnel-shaped soap dispenser [18]



This reference design changed the geometry of the bottle to facilitate efficient utilization of the lotion by enabling complete drainage toward the pump inlet. The bottom of the container is engineered with a funnel-shaped, allowing residual lotion to naturally flow toward the lowest point where the suction tube is located. This design achieves the goal of minimal product waste and consistent dispensing performance throughout the bottle's lifespan.

However, bottles shaped in this way compromise stability to an extent, and they may be more prone to tipping over.

Appendix C: Empty to Filled Weight Ratio Limit

The limit for empty bottle weight was derived through research of the heaviest type of refillable soap dispensers on the market. Investigation revealed that glass bottles had the heaviest empty to filled ratio of 8.25%. A rounded value of 10% was chosen to provide a more practical design limit and accommodate manufacturing variability.

Example: Clear Glass Soap Dispenser with Plastic Pump [15]

Weight empty = 353 g

Volume capacity = 3900 mL

Density of hand soap = 1.007 g/cm³ [16]

$$\text{weight of filled bottle} = \text{density of soap} \times \text{volume} + \text{weight empty}$$

$$\text{weight of filled bottle} = 4280.3 \text{ g}$$

$$\frac{\text{weight of empty bottle}}{\text{weight of filled bottle}} = 8.25\%$$

Appendix B: Rescoped Needs, Goals, and Objectives

Goal 1: Functionality The design should dispense a majority of the fluid contained in a timely manner, while requiring minimal effort to use.	Obj 1: The dispensing mechanism should be able to extract $\geq 98\%$ of the product.	Obj 2: No more than 30N of force should be required to operate the design per dispense cycle.	Obj 3: The design should take no longer than 1 second to operate
Goal 2: Reliability The design should correctly dispense the target amount and function for longer than the expiration date of the substance.	Obj 1: Each dispense cycle should dispense within $\pm 10\%$ of the target dispense amount.	Obj 2: The design should be able to withstand a minimum of n cycles, where $n = (\text{volume of moisturizer in the container}) / (\text{volume of moisturizer dispensed each cycle})$	Obj 3: The design should remain functional for a minimum of 3 years.
Goal 3: Mobility The design should be freestanding and easy for users to move around.	Obj 1: Zero permanent installation attachments are required to operate.	Obj 2: The maximum total mass of the design shall not exceed 4.5kg	

Appendix C - Justifications for Requirements and Evaluation Criteria

Goals	Requirements	Justification
1: Functionality The design should dispense a majority of the fluid contained in a timely manner, while requiring minimal effort to use.	1: The dispensing mechanism should be able to extract $\geq 88.31\%$ of the product by volume.	Conventional soap bottles extract an average of 88.31% of the product inside before failing [4]. The design chosen should aim to extract more than that.
	2: No more than 30N of force should be required to operate the design per dispensation cycle.	ISO's recommendations suggest that the minimum comfortable pushing strength for diverse users (with one finger) is 40-60 N for male adults, and 30-60 N for female adults [5]. Since the use of a finger strongly reflects the action used to apply a force on a dispenser's actuator, taking the lowest boundary (30 N) ensures that the product is accessible to the widest range of individuals [5]. A study on the torque required to open jars proposed a limit of 1.8Nm [6]. The smallest containers for moisturizer have a diameter of 2.5 inches (0.0635m) [1]. Solving for the Force using the Torque formula outputs a maximum force of 28N. A tolerance of 2N had been added to this value to compensate for variance in more tightly sealed jars.
	3: The design should take no longer than 5 seconds to operate	In a study conducted on jar opening times in first-year engineering students, it took no longer than 4 seconds for all participants to open the jar [Appendix C]. A tolerance of 1 second has been added to this value.
2: Reliability The design should correctly dispense the target amount and function for longer than the expiration date of the substance.	1: Each dispensation cycle should dispense a volume of product within $\pm 18\%$ of the targeted amount.	According to a peer-reviewed study in the International Journal of Pharmaceutical Compounding, the average volume dispensed by users of a Dosing Pump Dispenser varied by 18% of its targeted amount [7].
	2: The design should be able to withstand a minimum of n cycles, where $n = (\text{volume of moisturizer in the container}) / (\text{volume of}$	According to section 672.314.2c of the Uniform Commercial Code: Sales, the expectation of merchandise is "Are fit for the ordinary purposes for which such goods are use[d]." In this scope, the merchant will be a dispenser, thus it should last as

	moisturizer dispensed each cycle)	long for its purpose of dispensing all the contents in the container [8].
	3: The design should remain functional for a minimum of 3 years.	The typical shelf life of an unopened moisturizer bottle is 3 years, with this number decreasing if the product has been opened. Accordingly, the design should be functional for at least 3 years [9].
3: Mobility The design should be freestanding and easy for users to move around.	1: Zero permanent installation attachments are required to operate the design.	There cannot be any permanent installation on the property because the stakeholders in the scope are people who rent or live in school residences.
	2: The mass of the design when containing none of product shall not exceed 2.5kg	According to ISO 11228-1:2021 Ergonomics - Manual handling - Part 1: Lifting, lowering and carrying, a risk need not be conducted when handling a product weighing less than 3kg [10] The heaviest moisturizers on the market for household sized moisturizer bottles weigh a maximum of 500g; thus, the weight of an empty design should not exceed 2.5kg [11]

Appendix D - Ideal Crater Shape

Although a crater is the chosen shape for the base of the proposed solution, the optimized geometry of this crater has to be empirically tested. To do this, a control shape (Base Case) was selected and then iterated upon in 8 different ways to test what qualities produce larger fluid correction.

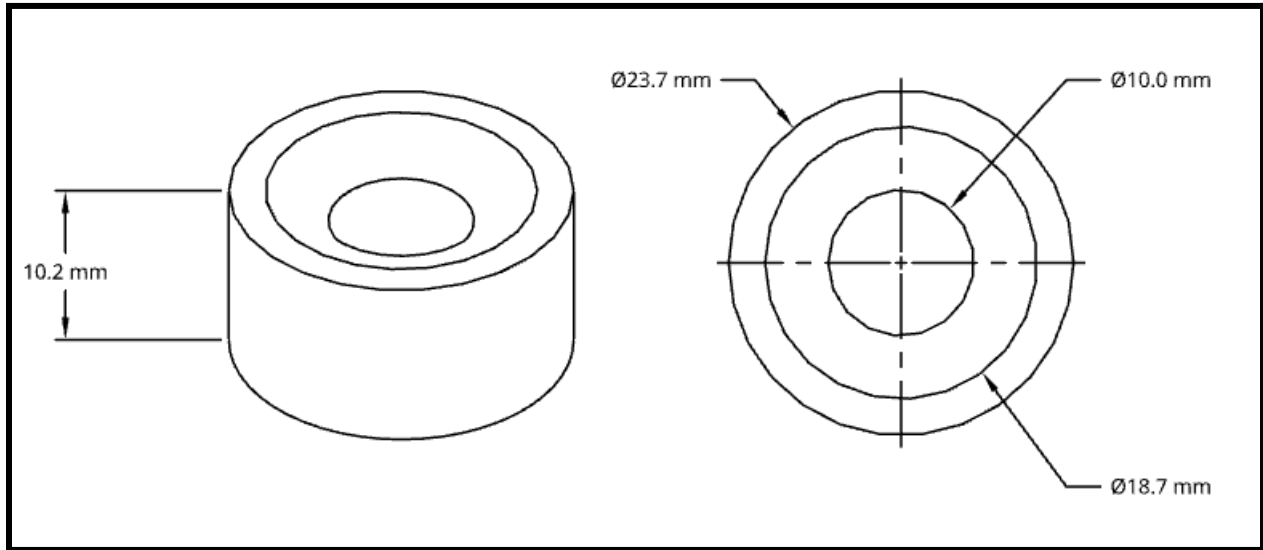


Figure 15: Crater control shape (Base Case).

Below is a table outlining the name and key dimension modifications used to test the performance of modifying the geometry of the Base Case.

Table 7: Name and Key Modifications of the 8 Craters Tested.

Name	Key Modification (mm)
C1: Base Case	Depth: 0.75
C2: Deep Funnel	Depth: 1.75
C3: Steep Walls	Radii: R0.25 / R0.25
C4: Gentle Slope	Radii: R1.0 / R1.0
C5: Funnel Profile	Upper: R1.0, Lower: R0.25
C6: Reverse Funnel	Upper: R0.25, Lower: R1.0
C7: Very Steep	Radii: R0.1 / R0.1
C8: Two Tier	Upper tier: 0.6 deep, R0.5 Lower tier: 0.65 deep, R0.3

To test these 8 different crater shapes against the control (Base Case), moisturizer was dispensed at the same location relative to the center of the respective crater shape. A picture was taken with a topview. 8 hours later, another picture was taken from the same angle. Using computer vision code, the displacement of the fluids relative to their center was calculated. This was done using a Python Script and the OpenCV library to track the displacement of the white pixels per crater shape [19].



Figure 16: Before, After, and Computer Vision Masking Pictures in the Crater Experiment

After applying the code, the following results were found, highlighting that C7: Very Steep was the highest performing crater with 10.68 pixels of moisturizer displaced.

Table 8: Name and Amount of Moisturizer Displaced of the 8 Craters Tested.

Name	Moisturizer Displaced (px)
C1: Base Case	0.0
C2: Deep Funnel	2.98
C3: Steep Walls	2.88
C4: Gentle Slope	1.24
C5: Funnel Profile	2.83
C6: Reverse Funnel	1.44
C7: Very Steep	10.68
C8: Two Tier	2.58

Empirically, the quantified results align with human observation. Thus, it can be successfully concluded that a crater shape influenced by C7: Very Steep.

Appendix E - Mean Jar Opening Times in First-year Engineering Students:

Participants were given a jar of CeraVe Moisturizing Cream and asked to twist off the lid at the pace they typically would when using the product. Each participant was asked to perform this task 3 times and were given a break of 30 seconds to recover between each trial. Times were recorded by an observer using the stopwatch function on the *Clock* app on an iPhone.

Table 9: Mean Jar Opening Times in First-year Engineering Students

	Time Taken (s)	Mean Time Taken (s)
Participant A	3.27	3.58
	3.64	
	3.83	
Participant B	3.33	3.35
	3.46	
	3.25	
Participant C	3.59	3.60
	3.76	
	3.45	
Participant D	3.64	3.96
	4.25	
	4.01	

Table E1: Time taken to open a jar of CeraVe Moisturizing Cream across 3 trials and 4 participants. The mean time taken can be found in the column on the right.

Appendix F - Moisturizer Gained From the Usage of the Transformable Pump/Jar:

The average commercial lotion bottle has a volume between 100-250mL or 300-500mL [11]. Thus, the residue in an average container is approximately 23.6ml ($0.082 * ((100+250)/2 + (300+500)/2) / 2$). According to a study on *exposure data for cosmetic products*, the median usage per day for lotion is 7.63g which equates to 7.63mL due to lotion's density of 1g/mL [20], [21]. Therefore, the extracted residue can last year for approximately an additional 3.1 days ($23.6 / 7.63$).

Appendix G: Video Link of Recommended Design

Video Link:

https://drive.google.com/file/d/1IHBYwzyczMs8CDCd_nUCPtm9cW8NbtM/view?usp=sharing

Appendix H: Source Extracts

[1] Common forms of packaging for moisturizers and their volume capacities



[2] Research on the Force used to operate squeeze bottles

The red line shows the approximate average (35N) of the force required to operate the squeeze bottle for the B-shampoo and D-shampoo products.

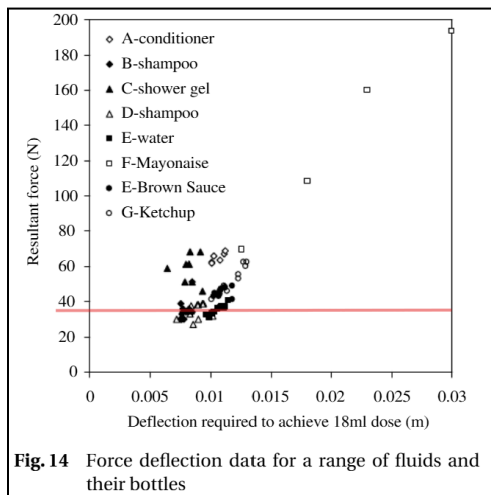


Fig. 14 Force deflection data for a range of fluids and their bottles

[3] Percentage of first-year engineering students who live in residence

Residence

Whether you are from overseas, another part of Canada or the Toronto area, residence is a great way to be part of the U of T community. Each year, approximately half of our first-year Engineering students live in residence. Engineering students lead busy lives, and living on campus gives you full access to academic resources, student services, and plenty of friends.

[9] Shelf life of moisturizer

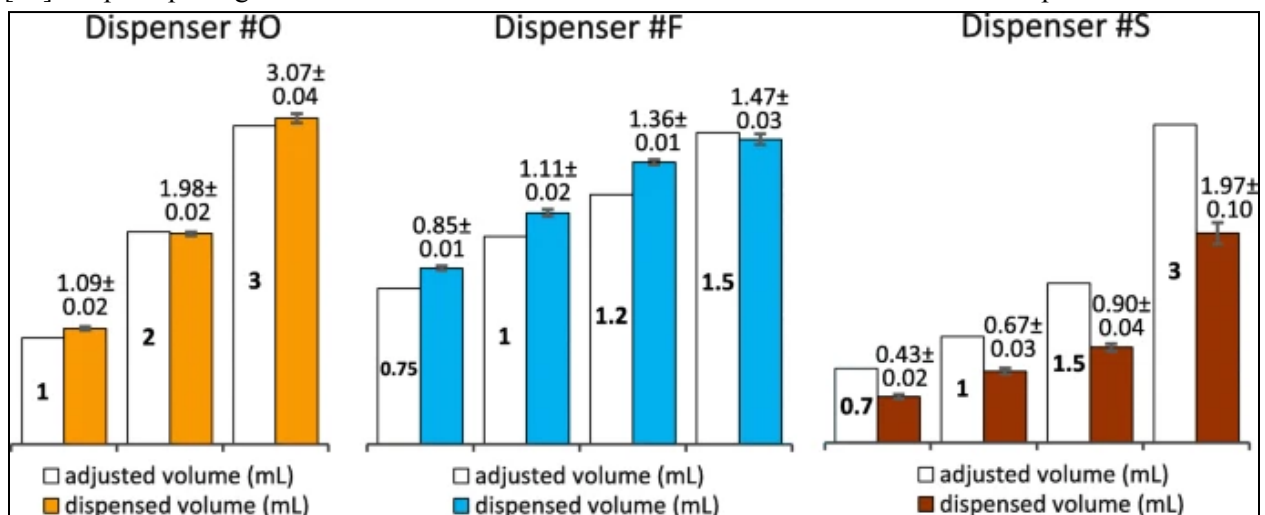
When it comes to body moisturizer, throw away unfinished containers that have been open for more than two years, regardless of their smell or texture. However, if you're using a lotion on a regular basis, odds are you'll need to replenish it before it expires. Sealed and unopened bottles should be good for three years.

[10] ISO 11228-1:2021- Ergonomics - Manual handling - Part 1: Lifting, lowering and carrying

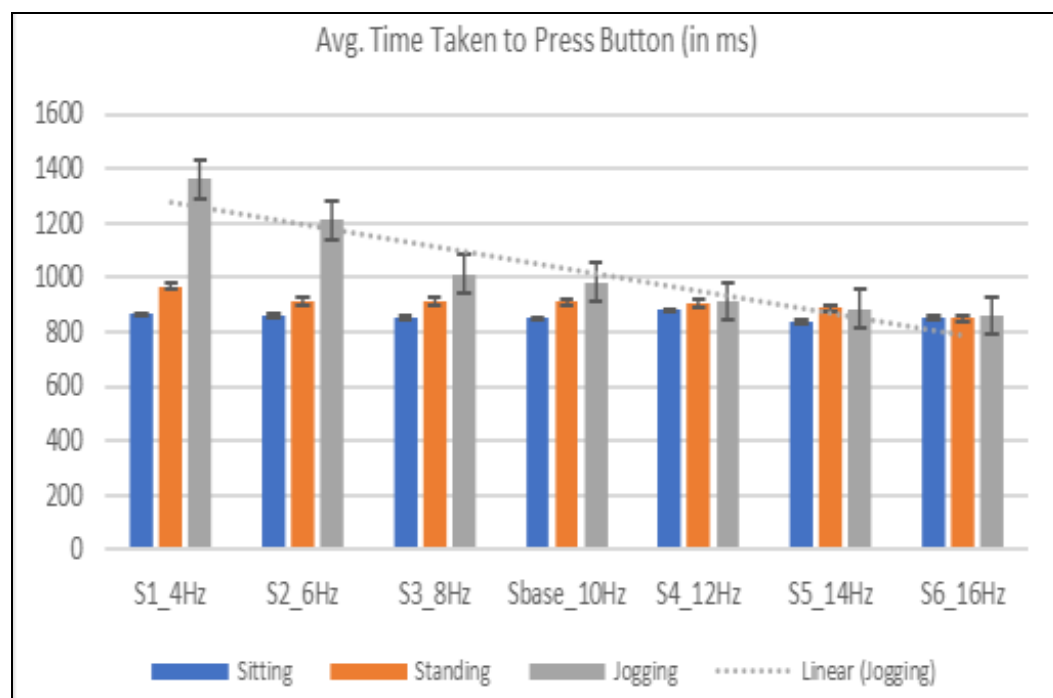
4.2.1 Using the step model

The step model illustrated in [Figure 1](#) describes the steps involved in beginning, and working through, a risk assessment of manual handling tasks, including lifting and carrying. Initially, the mass of the object being handled is determined; if it is more than 3 kg, the risk assessment is continued. The task is further analysed to determine if the mass exceeds recommended limits for handling (step 1).

[13] Graph depicting the variance between the desired amount and the actual amount dispensed



[18] Average Time taken to Press a Button in ms



9. Citations

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